HEALTHCARE ANALYTICS

Power BI Dashboard Documentation

Professional Project Report
Data Source: Healthcare Analytics Patient Dataset
Dashboard Platform: Microsoft Power BI

Prepared for Project / Portfolio Documentation

1. Executive Summary

This document provides professional documentation for the Healthcare Analytics Power BI dashboard. The underlying workbook contains 1,200 patient records across 23 fields. The dashboard is designed to transform patient-level healthcare data into interactive management insights, enabling users to explore patient demographics, clinical characteristics, healthcare utilization, and key performance indicators.

The report covers the dataset, data preparation, analytical approach, dashboard structure, key visualizations, business questions, interpretation guidelines, and recommended future enhancements.

2. Project Objectives

- Build an interactive healthcare analytics dashboard in Power BI.
- Convert raw patient-level records into clear, decision-oriented KPIs and visual insights.
- Analyze patient demographics and clinical attributes through filters and visual comparisons.
- Support identification of patterns, outliers, and areas requiring further investigation.
- Provide a reusable reporting framework for healthcare analytics and management review.

3. Dataset Overview

Metric	Value	Description
Patient records	1,200	Rows available in the source workbook
Data fields	23	Columns in the cleaned dataset
Source format	Excel (.xlsx)	Cleaned healthcare analytics workbook
Dashboard	Power BI (.pbix)	Interactive visualization and analysis layer

4. Data Dictionary / Field Inventory

Field	Distinct Values	Analytical Use

Patient_ID	1200	Identifier / record-level reference
Patient_Name	296	Categorical segmentation / filtering
Age	89	Numerical analysis / KPI / aggregation
Gender	2	Categorical segmentation / filtering
Hospital	5	Categorical segmentation / filtering
Department	8	Categorical segmentation / filtering
Doctor	10	Categorical segmentation / filtering
Diagnosis	14	Categorical segmentation / filtering
Admission_Date	670	Categorical segmentation / filtering
Discharge_Date	673	Categorical segmentation / filtering
Discharge_Date_Status	2	Categorical segmentation / filtering
Admission_Type	3	Categorical segmentation / filtering
Bed_Type	4	Categorical segmentation / filtering
Length_of_Stay	19	Numerical analysis / KPI / aggregation
Treatment_Cost	597	Numerical analysis / KPI / aggregation
Insurance_Provider	6	Categorical segmentation / filtering
Patient_Status	4	Categorical segmentation / filtering
Outcome	5	Categorical segmentation / filtering
Readmission	2	Categorical segmentation / filtering
Patient_Satisfaction	31	Numerical analysis / KPI / aggregation
Admission_Year	3	Numerical analysis / KPI / aggregation
Admission_Month	12	Numerical analysis / KPI / aggregation
Admission_Month_Name	12	Categorical segmentation / filtering

5. Data Preparation and Power BI Workflow

Import — Load the cleaned Excel workbook into Power BI Desktop.

Profile — Review field names, data types, blanks, duplicates, and value distributions.

Transform — Use Power Query to standardize data types, clean text values, handle missing values, and create analysis-ready fields where required.

Model — Create appropriate relationships and organize fields into logical analytical groups.

Measure — Create DAX measures for patient counts, percentages, averages, totals, and other KPIs required by the dashboard.

Visualize — Build cards, charts, slicers, tables, and supporting visuals with consistent formatting.

Validate — Cross-check dashboard totals against the source workbook and test slicers and interactions.

Publish — Publish the completed report to the Power BI Service when sharing and scheduled refresh are required.

6. Dashboard Structure

The uploaded PBIX file is retained as the authoritative dashboard artifact. The Word documentation describes the dashboard architecture and analysis rather than reproducing the PBIX canvas.

7. KPI and Visual Design Framework

- KPI cards: summarize high-level patient and healthcare metrics at a glance.
- Slicers: allow users to filter the dashboard by relevant demographic, clinical, or operational dimensions.
- Bar/column charts: compare categories and identify high/low segments.
- Donut/pie charts: show composition where a small number of categories makes proportions easy to read.
- Line charts: analyze trends where a valid date/time field is available.
- Tables/matrices: provide detailed record-level or grouped supporting information.
- Tooltips and cross-filtering: allow users to move from summary metrics to supporting detail.

8. Analytical Questions Addressed

1. How many patient records are represented in the dataset?
2. What are the major demographic and categorical segments?
3. How do patient characteristics differ across available groups?
4. Which categories contribute the largest or smallest shares of the population?
5. Are there visible patterns or outliers that warrant further investigation?
6. How can interactive filters be used to drill into specific patient segments?
7. Which KPIs are most useful for management-level monitoring?

9. Data Quality and Validation

Before operational use, the dashboard should be validated against the source workbook. Recommended checks include record counts, blank values, duplicate identifiers, category spelling, numeric ranges, date validity, and reconciliation of every major KPI with an independently calculated total.

Because healthcare data can be sensitive, access controls, minimum-necessary reporting, secure sharing, and appropriate anonymization should be applied before publishing or distributing the dashboard.

10. Recommended Enhancements

- Add a dedicated Executive Summary page with the most important KPIs and exception indicators.
- Add a Patient Demographics page for deeper segmentation.
- Add a Clinical / Utilization page for condition and service analysis where relevant fields exist.
- Add drill-through pages for detailed segment investigation.
- Add dynamic titles and KPI comparisons using DAX.

- Add a data-quality indicator showing the latest refresh and completeness status.
- Configure scheduled refresh and row-level security in Power BI Service where required.
- Document all DAX measures and business definitions in a separate measure catalog.

11. Conclusion

The Healthcare Analytics Power BI solution provides a structured approach to converting patient-level data into interactive analytical reporting. With appropriate validation, governance, and continued enhancement, the dashboard can serve as a practical reporting layer for healthcare analytics, monitoring, and exploratory decision support.

12. Project Artifacts

- Power BI dashboard: health care.pbix
- Source dataset: Healthcare_Analytics_1200_Patients_Cleaned(1).xlsx

13. Dashboard Screenshot

The following screenshot presents the completed Healthcare Analytics Power BI dashboard. It includes interactive filters for Hospital, Admission Year, and Department, together with summary KPIs and visual analyses covering patients by department, gender, stay group, diagnosis, hospital, treatment cost, and outcome.



Figure 1. Healthcare Analytics Dashboard – Power BI